EE1007
Introduction to Computer Science Laboratory

Lab 5: Control statement II
How to Debug

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How to debug

- When you write your code…
  - You must encounter bugs…





Outline



- Syntax error
- Logic error
- Debugger





Outline



- Syntax error
- Logic error
- Debugger



Syntax error

- Syntax error means the program cannot be successfully compiled and no executable file will be generated.
- Example:
  - Write a program to print the numbers from one to ten.

```
1 2 3 4 5 6 7 8 9 10

Process returned 0 (0x0)   execution time : 0.055 s
Press any key to continue.
```


Program cannot be successfully compiled

main.cpp

```
1  include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int counter = 1;
7
8      while(counter <= 10){
9          cout << counter << " ";
10         counter++;
11     }
12
13     cout << endl;
14     return 0;
15 }
```

Logs & others

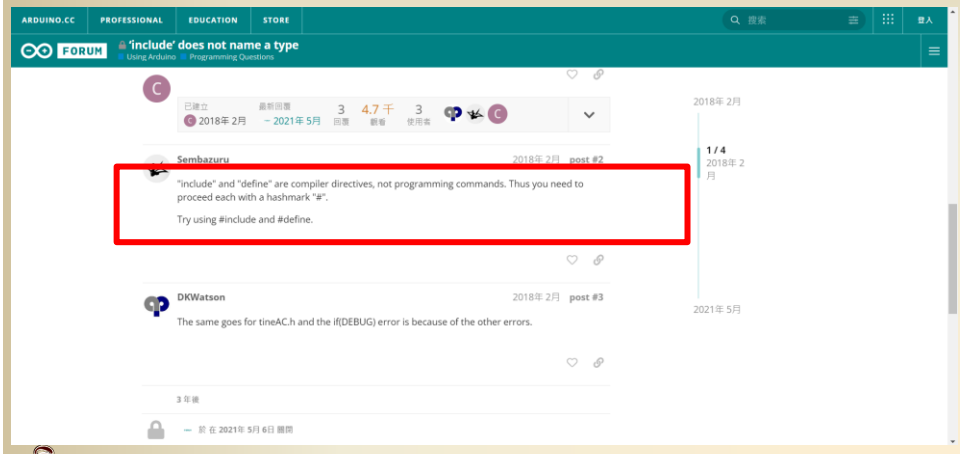
Code::Blocks | Search results | Cccc | Build log | Build messages

File	Line	Message
C:\Users\user...	1	error: 'include' does not name a type
C:\Users\user...		in function 'int main()':
C:\Users\user...	9	error: 'cout' was not declared in this scope
C:\Users\user...	9	note: suggested alternative: 'counter'
C:\Users\user...	13	error: 'cout' was not declared in this scope
C:\Users\user...	13	note: suggested alternative: 'counter'
C:\Users\user...	13	error: 'endl' was not declared in this scope
C:\Users\user...	13	note: suggested alternative: 'enum'
		=== Build failed: 4 error(s), 0 warning(s) (0 minute(s), 0 second(s)) ===

Search the meaning of error message



Program without



Correct the code

main.cpp

```
2   using namespace std;
3
4   int main()
5   {
6       int counter = 1;
7
8       while(counter <= 10){
9           cout << counter << " ";
10          counter++;
11      }
12
13      cout << endl;
14      return 0;
15  }
16
```

```
1 2 3 4 5 6 7 8 9 10

Process returned 0 (0x0)   execution time : 0.055 s
Press any key to continue.
```


Spell wrong

```
1   #includ <iostream>
2   using namespace std;
3
4   int main()
5   {
6       int counter = 1;
7
8       while(counter <= 10){
9           cout << counter << " ";
10          counter++;
11      }
12
13      cout << endl;
14      return 0;
15  }
```

Logs & others

Code::Blocks | Search results | Cccc | Build log | Build messages | CppCheck/V

File	Line	Message
C:\Users\user...	1	error: invalid preprocessing directive #includ; did you mean #include?
C:\Users\user...		
C:\Users\user...	9	error: 'cout' was not declared in this scope
C:\Users\user...	9	note: 'std::cout' is defined in header '<iostream>'; did you forget to '#include <iost...
C:\Users\user...	13	error: 'cout' was not declared in this scope
C:\Users\user...	13	note: 'std::cout' is defined in header '<iostream>'; did you forget to '#include <iost...
C:\Users\user...	13	error: 'endl' was not declared in this scope
C:\Users\user...	13	note: 'std::endl' is defined in header '<ostream>'; did you forget to '#include <ostre...
		=== Build failed: 4 error(s), 0 warning(s) (0 minute(s), 0 second(s)) ===
		=== Run: Debug in test (compiler: GNU GCC Compiler) ===

Without using namespace std;

main.cpp

```
1  #include <iostream>
2
3
4  int main()
5  {
6      int counter = 1;
7
8      while(counter <= 10){
9          cout << counter << " ";
10         counter++;
11     }
12
13     cout << endl;
14     return 0;
15 }
```

Logs & others

Code::Blocks | Search results | Cccc | Build log | Build messages | CppCheck/V

File	Line	Message
		=== Build: Debug in test (compiler: GNU GCC Compiler) ===
C:\Users\user...	9	error: 'cout' was not declared in this scope
C:\Users\user...	9	note: suggested alternative:
C:\Program Fi...	61	note: 'std::cout'
C:\Users\user...	13	error: 'cout' was not declared in this scope
C:\Users\user...	13	note: suggested alternative:
C:\Program Fi...	61	note: 'std::cout'
C:\Users\user...	13	error: 'endl' was not declared in this scope
C:\Users\user...	13	note: suggested alternative:
C:\Program Fi...	590	note: 'std::endl'
		=== Build failed: 3 error(s), 0 warning(s) (0 minute(s), 0 second(s)) ===

Without ；

main.cpp

```
2  using namespace std;
3
4  int main()
5  {
6      int counter = 1;
7
8      while(counter <= 10){
9          cout << counter << " ";
10         counter++;
11     }
12
13     cout << endl;
14     return 0
15 }
16
```

Logs & others

Code::Blocks | Search results | Cccc | Build log | Build messages

File	Line	Message
		=== Build: Debug in test (compiler: GNU GCC Compiler) ===
C:\Users\user...		In function 'int main()':
C:\Users\user...	14	error: expected ';' before '}' token
		=== Build failed: 1 error(s), 0 warning(s) (0 minute(s), 0 second(s)) ===

{ } didn't match

main.cpp

```
2  using namespace std;
3
4  int main()
5  {
6      int counter = 1;
7
8      while(counter <= 10){
9          cout << counter << " ";
10         counter++;
11
12
13     cout << endl;
14     return 0;
15 }
16
```

Logs & others

Code::Blocks | Search results | Cccc | Build log | Build messages

File	Line	Message
		=== Build: Debug in test (compiler: GNU GCC Compiler) ===
C:\Users\user...		In function 'int main()':
C:\Users\user...	15	error: expected '}' at end of input
C:\Users\user...	5	note: to match this '{'
		=== Build failed: 1 error(s), 0 warning(s) (0 minute(s), 0 second(s)) ===

Operator does not match

main.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int counter = 1;
7
8      while(counter <= 10){
9          cout >> counter >> " ";
10         counter++;
11     }
12
13     cout << endl;
14     return 0;
15 }
```

Logs & others

Code::Blocks | Search results | Cccc | Build log | Build messages | CppCheck/Ver

File	Line	Message
		=== Build: Debug in test (compiler: GNU GCC Compiler) ===
C:\Users\user...	9	error: no match for 'operator>>' (operand types are 'std::ostream' {aka 'std::basic_os...
C:\Users\user...	9	note: candidate: 'operator>>(int, int)' <built-in>
C:\Users\user...	9	note: no known conversion for argument 1 from 'std::ostream' {aka 'std::basic_ostrea...
C:\Program Fi...	1465	note: candidate: 'template<class _CharT, class _Traits, class _Alloc> std::basic_istre...
C:\Program Fi...	1465	note: template argument deduction/substitution failed:
C:\Users\user...	9	note: 'std::ostream' {aka 'std::basic_ostream<char>'} is not derived from 'std::basi...
C:\Program Fi...	931	note: candidate: 'template<class _CharT, class _Traits> std::basic_istream<_CharT, _Tr...
C:\Program Fi...	931	note: template argument deduction/substitution failed:
C:\Users\user...	9	note: 'std::ostream' {aka 'std::basic_ostream<char>'} is not derived from 'std::basi...
C:\Program Fi...	756	note: candidate: 'template<class _Traits> std::basic_istream<char, _Traits>& std::oper...
C:\Program Fi...	756	note: template argument deduction/substitution failed:
C:\Users\user...	9	note: 'std::ostream' {aka 'std::basic_ostream<char>'} is not derived from 'std::basi...
C:\Program Fi...	761	note: candidate: 'template<class _Traits> std::basic_istream<char, _Traits>& std::oper...
C:\Program Fi...	761	note: template argument deduction/substitution failed:

Without declared variables

main.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6
7      while(counter <= 10){
8          cout << counter << " ";
9          counter++;
10     }
11
12     cout << endl;
13     return 0;
14 }
```

Logs & others

Code::Blocks | Search results | Cccc | Build log | Build messages

File	Line	Message
		=== Build: Debug in test (compiler: GNU GCC Compiler) ===
C:\Users\user...		In function 'int main()':
C:\Users\user...	7	error: 'counter' was not declared in this scope
C:\Users\user...	7	note: suggested alternative: 'toupper'
		=== Build failed: 1 error(s), 0 warning(s) (0 minute(s), 0 second(s)) ===

Outline

- Syntax error
- Logic error
- Debugger

Logic error

- Logic error means the result of program isn't correct.
  - Usually the executable file is generated but not worked correctly.
- Example:
  - We use the same example mentioned above.

```
1 2 3 4 5 6 7 8 9 10

Process returned 0 (0x0)   execution time : 0.055 s
Press any key to continue.
```


Results of the program

- Miss 10

```
1 2 3 4 5 6 7 8 9

Process returned 0 (0x0)   execution time : 0.021 s
Press any key to continue.
```

Results of program

◆Check the setting of conditions in your code

main.cpp

```
#include <iostream>
using namespace std;

int main()
{
    int counter = 1;
    while(counter < 10){
        cout << counter << " ";
        counter++;
    }

    cout << endl;
    return 0;
}
```


Results of program

◆Find out the error

main.cpp

```
#include <iostream>
using namespace std;

int main()
{
    int counter = 1;
    while(counter < 10)
        cout << counter << " ";
        counter++;
    }

    cout << endl;
    return 0;
}
```

Results of program

- Correct your code

main.cpp

```
#include <iostream>
using namespace std;

int main()
{
    int counter = 1;
    while(counter <= 10){
        cout << counter << " ";
        counter++;
    }

    cout << endl;
    return 0;
}
```

```
C:\Users\user\Desktop\test\bin\Debug\test.exe
1 2 3 4 5 6 7 8 9 10

Process returned 0 (0x0)   execution time : 0.020 s
Press any key to continue.
```


Debug by cout

- Example: Find the maximum power of basis number under specific limit.

```
basis number=10
limit=5000
Maximum number is 10^3

Process returned 0 (0x0)   execution time : 7.763 s
Press any key to continue.
```

Debug by cout

- Logic error leads the program stuck in the loop.

main.cpp

```
#include <iostream>
using namespace std;

int main()
{
    float num, power, limit,result;
    int counter;

    cout << "basis number=";
    cin >> num;
    cout <<"limit=";
    cin >> limit;
    result = num;
    power = num;
    counter = 0;
    while(result < limit){
        result = num * power;
        counter++;
    }
    cout << "Maximum number is " << num << "^" << counter  << endl;

    return 0;
}
```

```
C:\Users\user\Desktop\test\bin\Debug\test.exe
basis number=10
limit=5000
```


Debug by cout

- Add cout into the loop

main.cpp

```
#include <iostream>
using namespace std;

int main()
{
    float num, power, limit,result;
    int counter;

    cout << "basis number=";
    cin >> num;
    cout <<"limit=";
    cin >> limit;
    result = num;
    power = num;
    counter = 0;
    while(result < limit){
        result = num * power;
        cout << result << endl;
        counter++;
    }
    cout << "Maximum number is " << num << "^" << counter  << endl;

    return 0;
}
```

```
basis number=10
limit=5000
100
100
100
```

Debug by cout

- Find the logic error

*main.cpp

```
#include <iostream>
using namespace std;

int main()
{
    float num, power, limit,result;
    int counter;

    cout << "basis number=";
    cin >> num;
    cout <<"limit=";
    cin >> limit;
    result = num;
    power = num;
    counter = 0;
    while(result < limit){
         result = result * power;
         //cout << result << endl;
         counter++;
    }
    cout << "Maximum number is " << num << "^" << counter  << endl;

    return 0;
}
```

```
basis number=10
limit=5000
Maximum number is 10^3

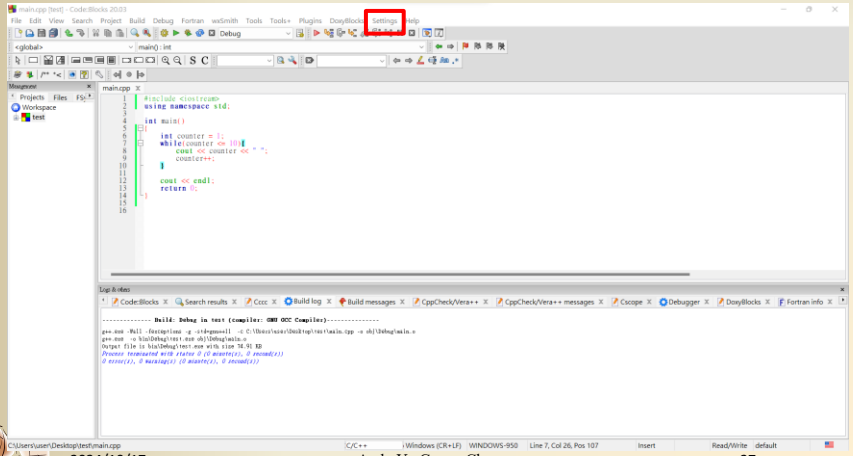
Process returned 0 (0x0)   execution time : 7.763 s
Press any key to continue.
```


Outline

- Sentence error
- Logic error
- Debugger

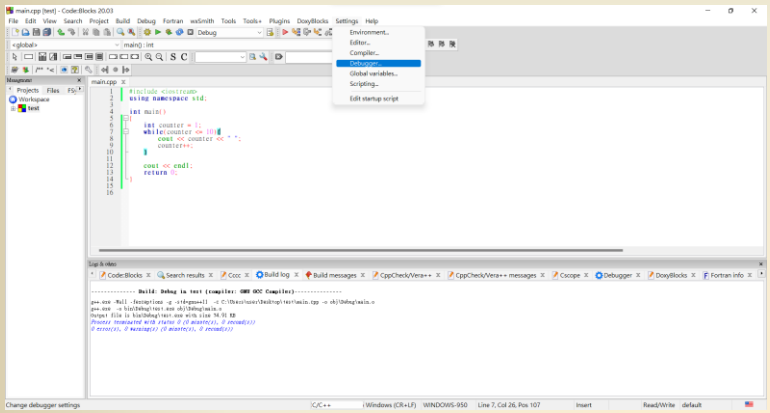
Debugger mode setting

◆Press “setting”



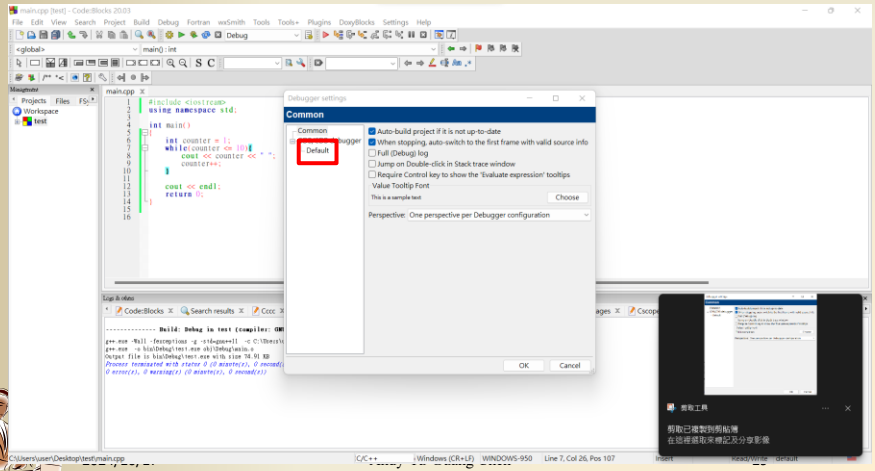
Debugger mode setting

◆Press “debugger…”



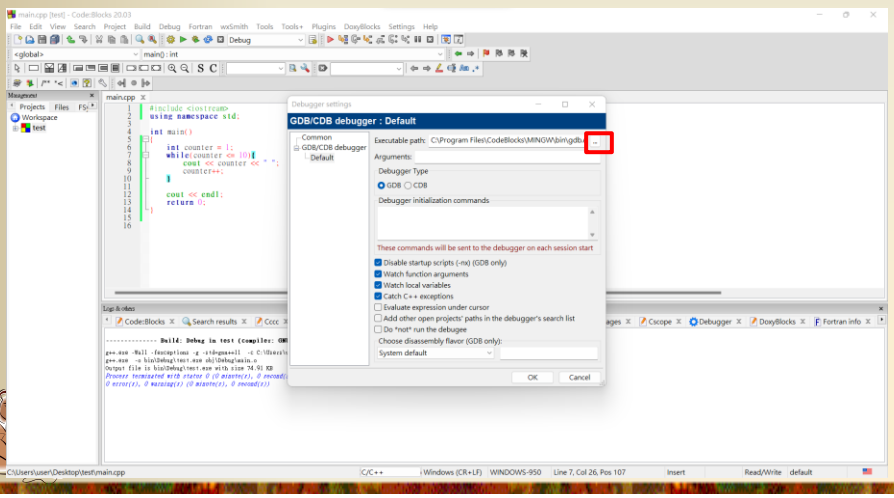
Debugger mode setting

◆Press “Default”



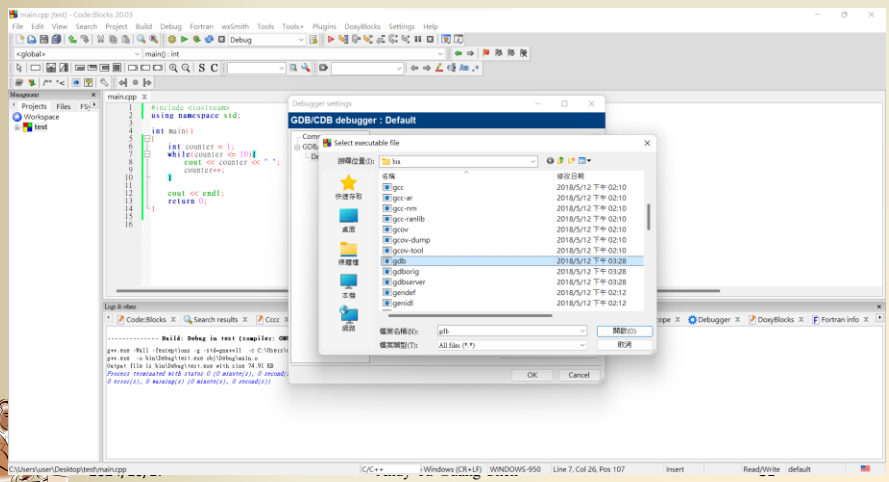
Debugger mode setting

◆Change executable path



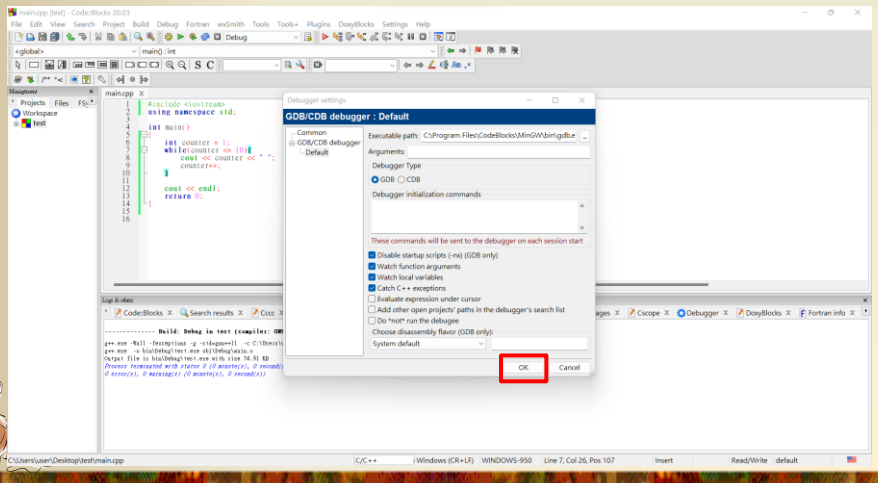
Debugger mode setting

◆ Choose C:\Program Files (x86)\CodeBlocks\MinGW\bin\gdb



Debugger mode setting

◆Press “OK”



Debugger mode

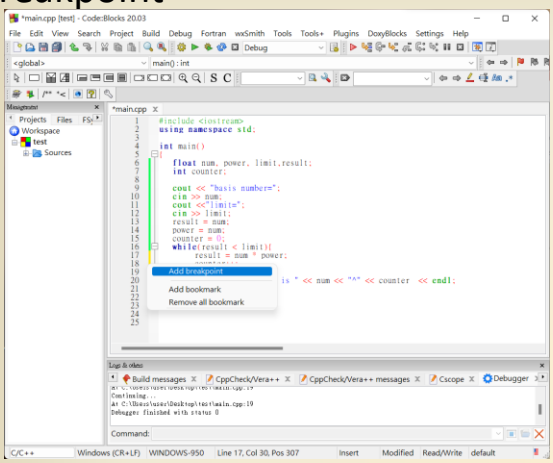
◆Add breakpoint

*main.cpp

```
1   #include <iostream>
2   using namespace std;
3
4   int main()
5   {
6       float num, power, limit,result;
7       int counter;
8
9       cout << "basis number=";
10      cin >> num;
11      cout <<"limit=";
12      cin >> limit;
13      result = num;
14      power = num;
15      counter = 0;
16      while(result < limit){
17            result = num * power;
18            counter++;
19      }
20      cout << "Maximum number is " << num << "^" << counter  << endl;
21
22      return 0;
23  }
```

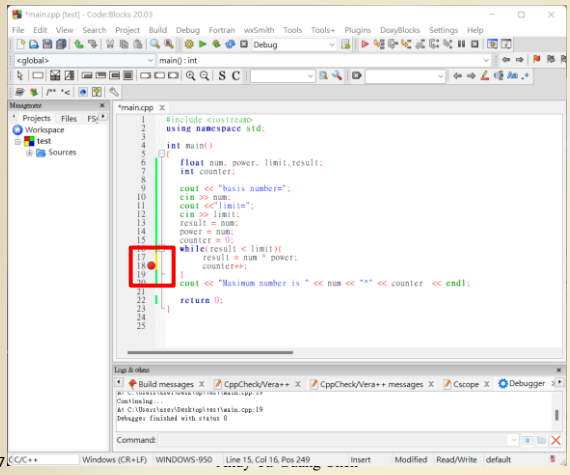

Debugger mode

◆Add breakpoint



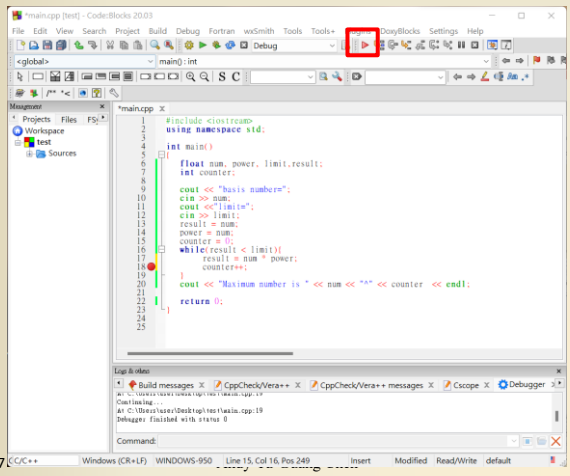
Debugger mode

- Add breakpoint



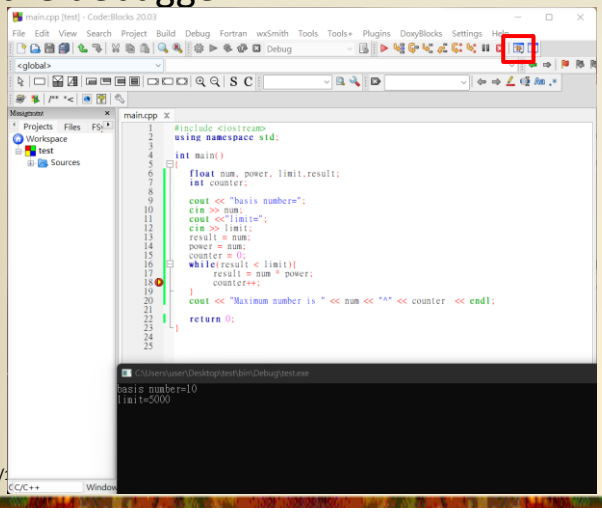
Debugger mode

- Run the debugger



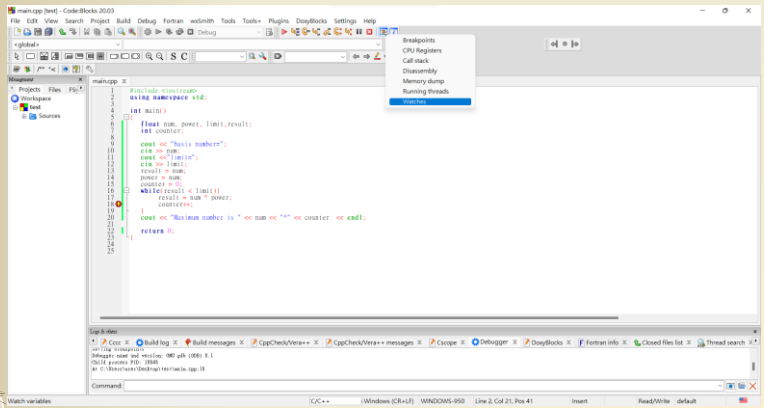
Debugger mode

- Run the debugger



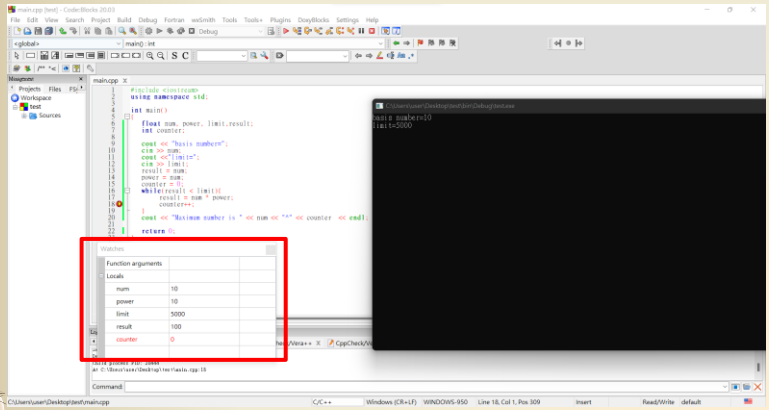
Debugger mode

- Run the debugger



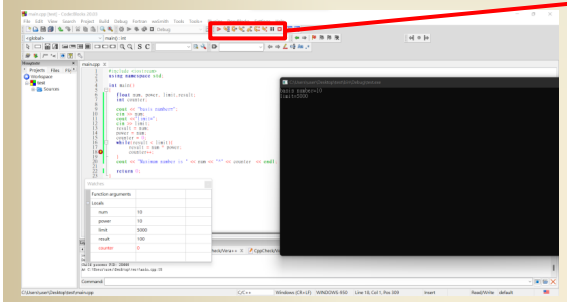
Debugger mode

- Run the debugger



Debugger mode

- Run the debugger





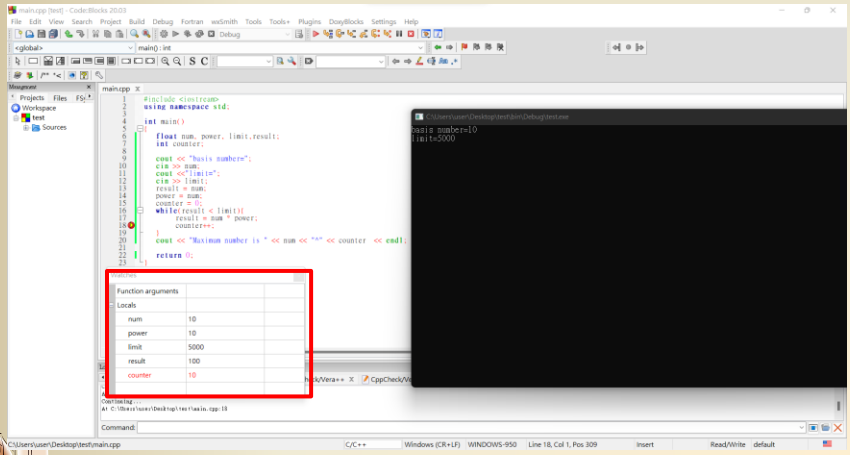
Run to cursor (F4):
Jump to the next break point.



Next line (F7):
Jump to the next line.

Debugger mode

◆Run the debugger



Debug by cout

◆Find the logic error

```
*main.cpp
#include <iostream>
using namespace std;

int main()
{
    float num, power, limit,result;
    int counter;

    cout << "basis number=";
    cin >> num;
    cout <<"limit=";
    cin >> limit;
    result = num;
    power = num;
    counter = 0;
    while(result < limit){
        result = result * power;
        //cout << result << endl;
        counter++;
    }
    cout << "Maximum number is " << num << "^" << counter  << endl;

    return 0;
}
```

```
basis number=10
limit=5000
Maximum number is 10^3

Process returned 0 (0x0)   execution time : 7.763 s
Press any key to continue.
```